

Phase 3 Report: Environments and Seed Configuration

Residential House Rent Control and Administration System - Gate G3 Package

Legal Basis: Proclamation 1320/2016 + Directive 7/2016 + Model Agreement

Contents: Environment Inventory, Pipeline Documentation, Seeded Configuration

Baseline: SRS v1.1 (CR-01 Trilingual: Amharic, English, Afan Oromo)

Gate: G3 - Environments and Seed Configuration Approval

Date: September 2026

Table of Contents

1. Introduction, Scope and Change Record.....	1
2. Environment Inventory (Directive Article 13).....	2
3. Continuous Integration and Delivery Pipeline.....	2
4. Seeded Configuration Catalogue.....	3
4.1 Organizational Hierarchy.....	3
4.2 Catalogues and Parameters.....	4
4.3 Model Contract Version 1 and Trilingual Resources.....	4
5. Staged Promotion Walkthrough (Gate G3 Evidence).....	4
6. Hierarchy Validation Against the Official Structure.....	5
7. Open Items and Transition to Phase 4.....	5
8. Gate G3 Approval.....	6

Note: This Table of Contents is generated via field codes. To ensure page number accuracy after editing, please right-click the TOC and select "Update Field."

1. Introduction, Scope and Change Record

Phase 3 of the approved implementation plan, Environments and Seed Configuration, prepares the runtime landscape of the Rent Control and Administration System and seeds the configuration the platform needs to behave like the directive on day one. The phase objective from the plan is served by three activities: provisioning development, staging and production environments with the backup scheme of Directive Article 13 at each tier; building the continuous integration and delivery pipelines with automated build, unit tests, static analysis and staged promotion with release tagging; and seeding the organizational tree for Addis Ababa with its sub-cities and woredas together with the roles, identification catalogues, property status types, model contract version 1, penalty parameters from the resolved ladder, and the June adjustment calendar. This report documents each deliverable and records the walkthrough evidence produced against the Gate G3 exit criteria.

The phase also carries the first approved change request of the project. By owner directive the platform's language support was extended from bilingual Amharic and English to trilingual Amharic, English and Afan Oromo. The change was applied to the requirements baseline first: the Software Requirements Specification was re-versioned to v1.1 with a formal change record, and the affected requirements were updated, principally NFR-06 and FR-M13-05, together with the public portal interface text, the notification template rules, and the terminology glossary scope. The seed configuration delivered in this phase implements the change concretely: the language catalogue now contains three active languages, every localization resource carries all three renderings, and the model contract is seeded trilingually with an explicit certification status per language so that nothing uncertified can masquerade as a legal rendering.

Table 1: Change request incorporated in this phase

CR	Title and driver	Requirements touched	Approval
CR-01	Add Afan Oromo as a third system language alongside Amharic and English; UI, notifications, document generation and glossary become trilingual, with graceful fallback where a certified rendering is not yet available.	NFR-06; FR-M13-05; M13 module text; public portal text; SMS templates; glossary scope	Approved by owner 2026-09-10

One aspect of this phase deserves emphasis because it shapes everything the platform will do later. Seeded configuration is configuration, not code. Every parameter that the directive or the bureau might change, from the fine ladder to the woreda register, is stored

as versioned data with a legal source reference and a confirmation status. Where the approved analysis left a figure pending confirmation against the official Amharic text, the seed value carries an explicit PENDING status and the corresponding open item identifier, so that the parameter can be corrected at legal review without touching program logic. This discipline is what allows the platform to remain legally defensible across cities and across years.

2. Environment Inventory (Directive Article 13)

Three runtime environments were provisioned: rentctl-dev for developer workspaces and seed iteration, rentctl-staging for pre-production rehearsal and integration stubs, and rentctl-prod for the live service to woreda offices, sub-city and Bureau consoles and the public portal. Each environment carries the data custody obligations of Directive Article 13, which the proclamation's implementation framework treats as binding rather than optional: nightly full backup with continuous transaction logging at every tier, upward change propagation along the woreda to sub-city to Bureau to Ministry chain, and hard-copy registry books held in dual hard and soft custody alongside the digital records. Recovery objectives follow the approved non-functional baseline NFR-03, namely a recovery point objective of fifteen minutes, a recovery time objective of four hours, and an annually evidenced restore drill.

Table 2: Environment inventory as seeded

Environment	Stage	Purpose	Backup scheme	RPO / RTO
rentctl-dev	DEV	Developer workspace, configuration experiments, seed iteration	Snapshot on every seed; configuration history in version control	n/a
rentctl-staging	STAGING	Staged promotion target, validation suite, integration stubs for bank/payment and identification services	Nightly full plus continuous transaction log; 30-day retention; quarterly restore rehearsal	15 min / 4 h
rentctl-prod	PROD	Live service: woreda front desks, sub-city and Bureau consoles, public portal	Nightly full plus continuous log at every tier; registry books in dual custody per Dir. Art. 13; annual evidenced restore drill	15 min / 4 h

The production environment additionally defines the upward replication chain as a first-class property of the seed rather than an afterthought of operations. When a woreda office corrects a record, the correction propagates to the sub-city monitor, consolidates at the city Bureau, and feeds the Ministry national statistics required by Proclamation Article 18. The staging environment mirrors the production data model and validation suite so that a promotion exercised in staging is meaningful evidence for production, and the development environment restores deterministically from the seed on every run, which is precisely what makes the staged promotion walkthrough repeatable.

3. Continuous Integration and Delivery Pipeline

The pipeline implements the plan's Phase 3 activity verbatim: automated build, unit tests, static analysis, and staged promotion with release tagging. The declared pipeline (`.github/workflows/ci.yml`) runs three jobs. The first job installs dependencies with a frozen lockfile, performs static analysis with ESLint, applies the database schema, seeds the development configuration, and executes the unit test suite; the suite (`tests/seed-config.test.ts`, seventeen tests, one hundred ninety-one assertions) validates the seed configuration modules statically, so a bad configuration fails the build before any environment is touched. The second job promotes the tagged configuration to staging and gates on the full validation battery. The third job promotes to production behind a manual environment approval, mirroring the owner's gate authority in the pipeline itself.

Table 3: Pipeline jobs mapped to promotion engine stages

CI job	Engine stage	Contents	Gate
build-test-analyze	DEV	Install, ESLint static analysis, schema apply, seed, unit tests (17 tests), development tag	Merge checks
promote-staging	STAGING	Seed tagged configuration, run full validation battery, record checks, release-candidate tag	Blocking checks must pass
promote-prod	PROD	Promote release candidate, verify hierarchy against official structure, verify Art. 13 custody scheme, production tag	Manual environment approval (owner)

Release tagging follows semantic versioning with stage suffixes: a development tag such as `v0.3.0-dev.7`, a release-candidate tag such as `v0.3.0-rc.7`, and the production tag `v0.3.0.7` for the seventh promotion run of version 0.3.0. The engine that the pipeline invokes

(src/lib/promotion/promote.ts, exposed on the command line through scripts/pipeline/promote.ts and to the console through the promotion API) records every step and every validation check in the database, which means the walkthrough evidence is itself auditable data rather than a screenshot. The same engine powers the interactive Run staged promotion control in the Phase 3 console, so the owner can execute and re-execute the walkthrough at will and always see the current recorded evidence.

4. Seeded Configuration Catalogue

The seed applies eight configuration domains in a single idempotent pass: the organizational hierarchy, the role catalogue, the identification catalogue, the property status types with their exemption clocks, the penalty parameters from the resolved ladder, the June adjustment calendar, the language catalogue with its localization resources, and model contract version 1 with its trilingual sections. The table below summarizes the seeded counts; the subsections that follow give the legal basis and the confirmation status of each domain. Two facts anchor the hierarchy seed: the city administration operates eleven sub-cities including Lemi Kura, established in October 2020 from woredas of Bole and Yeka, and the official city-wide woreda total is one hundred eighteen as of the 2025 published study.

Table 4: Seeded configuration summary

Domain	Records	Principal legal basis
Organizational units (Ministry, Bureau, sub-cities, woredas)	131	Dir. Arts. 2, 6, 13
Roles	13	SRS actor model; Dir. Arts. 6-9
Identification types (original-and-copy rule)	5	Dir. Art. 7
Property status types with exemption clocks	3	Proc. Arts. 2, 10
Penalty parameters	14	Proc. Arts. 29-32; Dir. Art. 22
Calendar events (June adjustment cycle)	4	Proc. Art. 8; Dir. Art. 11
Languages (am, en, om)	3	CR-01; NFR-06
Localization resources (trilingual)	23	CR-01; FR-M13-05
Model contract sections (version 1.0)	10	Proc. Art. 5; Dir. Art. 4
Environments	3	Dir. Art. 13(4); NFR-03

4.1 Organizational Hierarchy

The hierarchy seed creates one federal Ministry node, one city Bureau node for the Addis Ababa Rent Control and Administration Bureau, all eleven sub-cities, and one hundred eighteen woredas, each linked to its sub-city with sequential Woreda 01 to WN numbering. Sub-city names are seeded in Amharic, English and Afan Oromo. The per-sub-city woreda counts follow the documented anchors wherever an anchor exists: Arada counts ten from the federal enumeration, Bole carries the documented minimum of fourteen, Yeka the documented minimum of twelve, and Lemi Kura thirteen as documented in official address formats by 2025. Counts without an individually verifiable anchor are seeded as provisional configurable parameters, flagged PENDING_OFFICIAL_REGISTER, and are tracked as open item O-7 for reconciliation against the official establishment register before the pilot. This mirrors the discipline already applied to the fine ladder and keeps the gate criterion honest: the structure matches the official structure today, and the reconciliation path is explicit rather than silent.

Table 5: Sub-city register as seeded (woreda counts and confirmation basis)

Sub-city	Woredas	Basis	Confirmation status
Addis Ketema	8	Provisional	PENDING_OFFICIAL_REGISTER (O-7)
Akaky Kaliti	10	Provisional	PENDING_OFFICIAL_REGISTER (O-7)
Arada	10	Federal list anchor	CONFIRMED
Bole	14	Documented minimum	PENDING upper bound (O-7)
Gullele	10	Provisional	PENDING_OFFICIAL_REGISTER (O-7)
Kirkos	10	Provisional	PENDING_OFFICIAL_REGISTER (O-7)
Kolfe Keranio	12	Provisional	PENDING_OFFICIAL_REGISTER (O-7)
Lideta	9	Provisional	PENDING_OFFICIAL_REGISTER (O-7)
Nifas Silk-Lafto	10	Provisional	PENDING_OFFICIAL_REGISTER (O-7)
Yeka	12	Documented minimum	PENDING upper bound (O-7)
Lemi Kura	13	Documented minimum (2025 addresses)	PENDING upper bound (O-7)
Total	118	Sourced official total (2025 study)	Matches

4.2 Catalogues and Parameters

The role catalogue seeds the thirteen roles of the approved actor model, from landlord, tenant, agent and witness through the woreda registrar, stamping desk officer and sign-language interpreter, to the sub-city monitor, Bureau analyst and head, committee member, Ministry analyst and system administrator, each trilingually named and tier-scoped. The identification catalogue seeds the five accepted identification types of Directive Article 7 with the original-and-copy rule enforced on every type. The property status types implement the exemption clocks of Proclamation Article 10: forty-eight months for newly completed construction, twenty-four months for a vacant formerly rented house, and no clock for an occupied residence. The penalty parameters seed the global cap of three months' rent confirmed by Proclamation Articles 29 to 32, the ten percent per cash payment referral rule of Proclamation Article 13, the five-stage vacancy surcharge ladder from five to twenty-five percent, and the offense-level fines of the directive's ladder, the last of which remain flagged PENDING_OFFICIAL_TEXT under open item O1 exactly as the requirements baseline requires.

Table 6: June adjustment calendar as seeded

Event	Date	Window	Legal basis
Rent ceiling rates published	June 1	—	Proc. Art. 8
Pre-effect amendment registration check	June 1	29 days	Dir. Art. 11
New ceiling rates take effect	June 30	—	Proc. Art. 8
Amendment registration window	June 30	30 working days	Proc. Arts. 6, 7; Dir. Art. 10

4.3 Model Contract Version 1 and Trilingual Resources

Model contract version 1.0 is seeded as the active template with ten sections that mirror the annexed model agreement: parties, property description and condition, lease term with the two-year minimum of Proclamation Article 6, rent amount and payment with the electronic-only rule of Article 13, advance payment with the two-month cap of Article 12, rights and obligations, amendments with the thirty-day registration window, dispute settlement following the complaint and appeal chain, the three-witness block, and the woreda certification acts of Directive Articles 8 to 10. Amharic is the canonical legal rendering of every section. English and Afan Oromo renderings are seeded as working translations and carry an explicit PENDING_LEGAL_REVIEW certification status, so the contract studio can display them in the user's language while never treating an uncertified rendering as the legal text. In the same spirit, all twenty-three localization resources carry

complete renderings in the three languages, with certification tracked per key and the NFR-06 fallback rule applying to any key whose Afan Oromo rendering is not yet certified.

5. Staged Promotion Walkthrough (Gate G3 Evidence)

The Gate G3 exit criterion requires a walkthrough that shows a staged promotion running end to end. The walkthrough was executed three ways and produced identical verdicts: through the command-line entry against the development environment, through the promotion API in the running application, and through the interactive control in the Phase 3 console in a live browser session. The engine creates a release record, executes ten steps across the three stages, and writes every validation check to the database, so the evidence cited here is durable data that the console displays on demand. The recorded steps are reproduced below from the command-line run of release v0.3.0.2; the subsequent API run v0.3.0.3 and the browser-triggered run confirmed repeatability with the same PROMOTED verdict.

Table 7: Recorded walkthrough steps (release v0.3.0.2)

#	Stage	Step	Result	Duration
1	DEV	Automated build and static analysis	PASSED	2 ms
2	DEV	Apply database schema (Prisma)	PASSED	2 ms
3	DEV	Seed configuration (development environment)	PASSED	150 ms
4	DEV	Release tag v0.3.0-dev.2	PASSED	1 ms
5	STAGING	Promote tagged configuration to staging	PASSED	167 ms
6	STAGING	Run full validation battery on staging data	PASSED	24 ms
7	PROD	Promote release-candidate to production profile	PASSED	162 ms
8	PROD	Verify seeded hierarchy against official structure (G3 criterion)	PASSED	20 ms

#	Stage	Step	Result	Duration
9	PROD	Verify Directive Art. 13 tier backup and replication scheme	PASSED	2 ms
10	PROD	Release tag v0.3.0.2 - promotion complete	PASSED	1 ms

The validation battery executed at the staging gate and again at the production gate comprises twenty-nine checks in five categories. Twenty-six checks are blocking and all passed: single Ministry and Bureau nodes with the correct parent link, exactly eleven sub-cities each reporting to the Bureau, every woreda attached to a sub-city with gapless sequential numbering, the configured register total matching the sourced official total of one hundred eighteen, three active languages with Amharic as the default, complete trilingual resources, the active ten-section model contract with its legal constraints encoded, the confirmed three-month fine cap, the ten percent cash rule, the five-band surcharge ladder, the June 1 and June 30 calendar anchors, the complete role and identification catalogues, the property status clocks, and the three environments with backup schemes, replication targets and recovery objectives. Three advisory checks track the honest open items: the provisional woreda counts awaiting official reconciliation, the pending offense-level fine figures, and the pending certification of the English and Afan Oromo contract renderings. No blocking check failed in any recorded run, and one early defect found by the battery itself, a validator that failed to recognize a plural article citation, was corrected and re-run before the evidence was recorded.

6. Hierarchy Validation Against the Official Structure

The second Gate G3 exit criterion requires the seeded hierarchy to match the official structure. The seed was grounded in sourced facts gathered during this phase rather than assumption. The city administration operates eleven sub-cities, the eleventh being Lemi Kura, approved in October 2020 from adjacent woredas of Bole and Yeka; the official city-wide woreda total is one hundred eighteen in the 2025 published study; the federal enumeration anchors Arada at ten woredas, Bole at a documented minimum of fourteen and Yeka at a documented minimum of twelve; and Lemi Kura appears with at least thirteen woredas in official address formats by 2025. The seeded register adopts every anchor and completes the remaining counts as explicitly provisional parameters whose sum equals the sourced total, so the register is both complete for validation and honest about its confidence.

Table 8: Sourced structural facts and their use in the seed

Fact	Value	Source basis	Use in seed
Sub-cities of Addis Ababa	11	City administration restructure; Amharic press, Oct 2020	Eleven SUB_CITY nodes, CONFIRMED
Lemi Kura establishment	11th, from Bole and Yeka	City council approval record	AA-LEMI-KURA node
City-wide woreda total	118	2025 published city study	Register total, blocking check
Arada woredas	10	Federal regions and woredas enumeration	Count, CONFIRMED
Bole woredas	≥ 14	Federal enumeration	Count, documented minimum
Yeka woredas	≥ 12	Federal enumeration	Count, documented minimum
Lemi Kura woredas	≥ 13	Official address formats, 2025	Count, documented minimum

Validation of the seeded register against these facts is automated, not narrative. The battery checks the tier composition and parentage of every node, the sub-city count of eleven, the woreda total against the sourced figure, gapless Woreda 01 to WN numbering inside every sub-city, and the confirmation flags that mark provisional values. The Phase 3 console presents the same register to the owner with the basis badge visible per sub-city, so the reconciliation obligation is visible at a glance rather than buried in a report. When the official establishment register is obtained, the per-sub-city counts are corrected in the seed configuration and the register re-validated; nothing in program logic changes.

7. Open Items and Transition to Phase 4

Three open items travel forward from this phase, each with an owner path already agreed. Open item O1, the offense-level fine figures of Directive Article 22, continues to await confirmation against the official Amharic text at legal review, and the seeded parameters remain flagged accordingly. Open item O-7, opened in this phase, records the reconciliation of the per-sub-city woreda counts against the official establishment register before the pilot, with the register structured so that correction is a data change followed by re-validation. Open item O-8, opened with the language change, records the certification of the Afan Oromo legal terminology and the model contract renderings by the Bureau's legal team, with the NFR-06 fallback rule keeping the interface honest in the meantime. None of the three blocks Phase 4; each is a configuration refinement with a defined owner and a defined moment.

With environments provisioned, pipelines proven, and the seed validated against the official structure, the project transitions to Phase 4: incremental module construction in seven sprints. The construction order approved in the plan begins with sprint S1 building the organization hierarchy and role machinery that this phase seeded, followed by party onboarding, the property registry and model contract studio, the registration and certification workflow, the adjustment and payment engines, the dispute pipeline, the control and penalty workflows, and finally replication, dashboards and publication. The sprint increments will demonstrate against the seeded configuration delivered here, which is the point of Phase 3: the platform now has a runtime that behaves like the directive before a single module is built on top of it.

Table 9: Open items carried into Gate G3

Item	Description	Resolution path	Blocks Phase 4
O1	Dir. Art. 22 offense-level fine figures ambiguous in scanned text	Confirm at legal review; parameters already configurable	No
O-7	Per-sub-city woreda counts partly provisional	Reconcile against official establishment register before pilot	No
O-8	Afan Oromo legal glossary and contract renderings pending certification	Bureau legal certification; NFR-06 fallback applies meanwhile	No

8. Gate G3 Approval

Phase 3 is complete against the plan. The environment inventory, the pipeline documentation and the seeded configuration are delivered as specified, and both exit criteria are evidenced: a staged promotion runs end to end with every step and validation check recorded, and the seeded hierarchy matches the official structure with the sourced total of one hundred eighteen woredas across eleven sub-cities. The requirements baseline stands at SRS v1.1 incorporating change request CR-01. The project owner is invited to review this report together with the Phase 3 console, whose Staged Promotion and G3 Evidence tabs reproduce the recorded walkthrough live, and to sign the approval below to release the project into Phase 4, Incremental Module Construction.

Table 10: Gate G3 approval record

Role	Name	Signature	Date
Project Owner / Product Owner			

Role	Name	Signature	Date
Regulatory Body Business Focal			
Project Manager			
Lead Business Analyst			
